

Physics-Aware Contribution-Guided Pruning for Low-Complexity Sparse Volterra Equalization in 100-Gb/s IM/DD PAM4 Links

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Abstract We experimentally demonstrate a novel physics-aware, contribution-guided pruned Volterra equalization framework for distortion mitigation in a 100-Gb/s PAM4 IM/DD link over 4.2-km SMF, achieving ~92% complexity reduction versus full VE and ~36.5% reduction relative to baseline MP-VE, while delivering superior performance at the KP4-FEC limit. ©2026 The Author(s)

Introduction

The rapid growth of cloud computing, artificial intelligence (AI), and data-center interconnects is driving demand for high-capacity, energy-efficient optical links. IM/DD with multi-level PAM has emerged as a preferred short-reach solution due to its low cost and simplicity [1]–[3]. However, scaling to higher data rates per wavelength is fundamentally limited by electro-optic bandwidth constraints, nonlinear distortions, and equalization-induced noise enhancement [4]. Volterra nonlinear equalizers (VNLEs) effectively mitigate these impairments by capturing higher-order signal interactions [5]–[7], but their computational complexity grows combinatorically with nonlinear order and memory depth, limiting real-time implementation [6], [7]. To address this, prior work has focused on sparse Volterra equalization via pruning and regularization techniques [8]–[10], primarily relying on magnitude-based coefficient selection. While computationally efficient, such approaches implicitly assume that coefficient magnitude reflects term importance. In nonlinear optical systems, this assumption is often invalid, as the actual contribution of each Volterra term depends jointly on its coefficient and the statistical energy of its associated basis function. This observation motivates a shift from magnitude-based sparsification toward contribution-aware pruning, directly inspiring the proposed physics-aware contribution-guided framework.

In this work, we propose and experimentally demonstrate, for the first time, a physics-aware contribution-guided Volterra equalization framework (PCG-VE) for distortion mitigation in a 100-Gb/s PAM4 IM/DD link over 4.2-km SMF. Unlike conventional approaches, proposed PCG-VE quantifies each Volterra term by its true signal contribution, enabling precise identification of dominant nonlinear interactions while eliminating redundant components, resulting in a physically

grounded and highly robust sparsification strategy. Our experimental results demonstrate a remarkable outcome: proposed PCG-VE (Contribution Pruning, CP) achieves BER comparable to the fully connected VE (27/27/10 taps) with ~92% coefficient reduction and no penalty, while baseline MP-VE (Magnitude Pruning, MP) degrades at the same sparsity due to loss of critical nonlinear interactions. Even at 96% reduction, proposed PCG-VE remains below the KP4-FEC threshold, confirming its superior robustness. We reveal a fundamental BER–complexity trade-off by dynamically scaling complexity (multiplications per symbol) and identifying the minimum coefficients required for near-optimal performance. While increasing model size improves BER, it rapidly escalates computational cost. The proposed PCG-VE resolves this by retaining only the most significant coefficients those contributing most to the reconstructed signal energy thereby eliminating redundant nonlinear terms. As a result, proposed PCG-VE achieves ~92% complexity reduction versus full VE and ~36.5% reduction relative to baseline MP-VE, while delivering superior performance at the KP4-FEC limit, establishing an efficient and scalable nonlinear equalization framework for bandwidth-limited systems.

Operation Principle of PCG-VE Framework

The received signal $x[n]$ is expanded using Volterra basis functions $\phi_i^{(p)}(x[n]) = \prod_{j=1}^p x[n - i_j]$, capturing linear and nonlinear distortions through interactions among delayed input samples (Fig. 1). The Volterra output (up to third order) is compactly expressed as

$$y[n] = \sum_{p=1}^3 \sum_{i=0}^{M_p-1} \sum_{j=0}^{M_p-1} \sum_{k=0}^{M_p-1} w_i^{(p)} \phi_i^{(p)}(x[n]) \quad (1)$$

where p denotes the Volterra order, M_p the number of terms, and $w_i^{(p)}$ the corresponding coefficients. The optimal coefficient vector w^* is

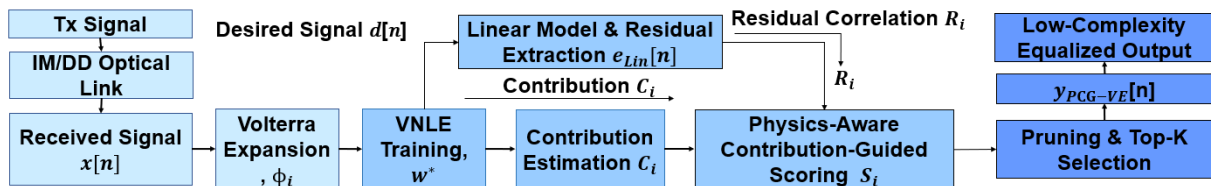


Fig. 1. Proposed PCG-VE framework for IM/DD links, where full Volterra training and residual-driven contribution estimation enable physics-aware pruning of dominant nonlinear terms, achieving low-complexity equalization with preserved BER performance.

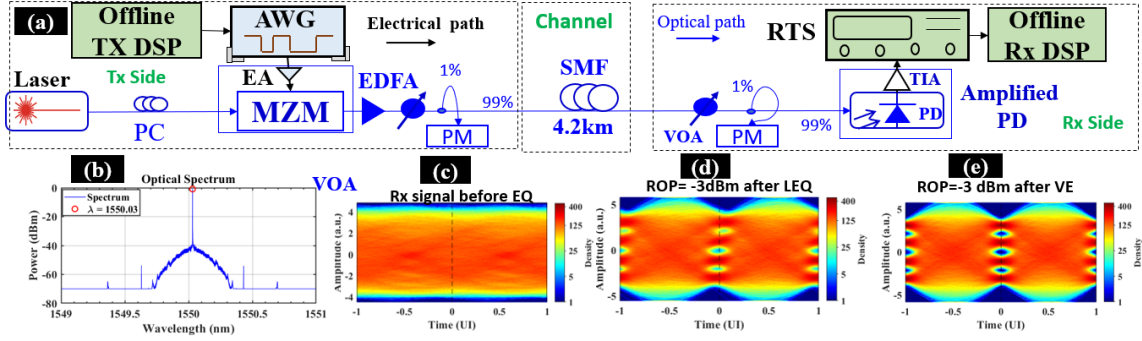


Fig. 2. (a) Experimental IM/DD optical link for validation under practical hardware constraints; (b) optical spectrum; (c) distorted received signal before equalization; (d) partial recovery using linear equalization (ROP = -3 dBm); and (e) enhanced recovery using nonlinear equalization (ROP = -3 dBm), highlighting bandwidth limitations and nonlinear distortions.

obtained by minimizing the mean-square error (MSE) between the desired signal $d[n]$ and the model output $y[n]$:

$$w^* = \arg \min_w \mathbb{E} [|d[n] - y[n]|^2] \quad (2)$$

where $\mathbb{E}[\cdot]$ denotes expectation and w stacks the first-, second-, and third-order coefficients. As shown in Fig. 1, the proposed method preserves all linear Volterra terms while pruning higher-order nonlinear components. The linear model is

$$y_{\text{Lin}}[n] = \sum_{i=1}^{M_1} w_i^{(1)} \phi_i^{(1)}(x[n]) \quad (3)$$

and the corresponding residual error is defined as

$$e_{\text{Lin}}[n] = d[n] - y_{\text{Lin}}[n] \quad (4)$$

which captures nonlinear distortion not explained by the linear model, guiding identification of dominant nonlinear terms. A physics-aware contribution metric is defined for each Volterra basis

$$C_i = |w_i|^2 \mathbb{E} [|\phi_i^{(p)}(x[n])|^2] \quad (5)$$

which quantifies the actual output energy contribution of each term. To further emphasize terms that remain relevant to the residual nonlinear distortion, a residual-correlation factor is introduced:

$$R_i = |\text{corr}(\phi_i^{(p)}(x[n]), e_{\text{Lin}}[n])| \quad (6)$$

Where $\text{corr}(\cdot)$ denotes correlation. Combining both metrics, the final pruning score (physics-aware block) is defined as

$$S_i = \begin{cases} C_i, & 1 \leq i \leq M_1 \\ C_i(1 + \alpha R_i), & M_1 + 1 \leq i \leq M \end{cases} \quad (7)$$

Thus, all linear terms (M_1) are preserved, while nonlinear terms ($M = M_2 + M_3$) are ranked by contribution strength and residual relevance, where α controls residual influence (data-adaptive). Given a sparsity level $\rho \in [0, 1]$, the number of retained nonlinear terms is $K_{\text{NL}} = \lfloor \rho M \rfloor$, where $\lfloor \cdot \rfloor$ denotes rounding. The top- K_{NL} nonlinear terms

are selected by sorting the scores S_i in descending order. Let $S_{\text{Lin}} = \{1, 2, 3, \dots, M_1\}$ denote the preserved linear set, and S_{NL} the selected nonlinear terms. The final active set is $S = S_{\text{Lin}} \cup S_{\text{NL}}$, and the resulting PCG-VE is expressed as

$$y_{\text{PCG-VE}}[n] = \sum_{p=1}^3 \sum_{i \in S} \sum_{j \in S} \sum_{k \in S} w_i^{(p)} \phi_i^{(p)}(x[n]) \quad (8)$$

This framework enables interpretable sparsification by preserving essential interactions while removing redundancy, achieving significant complexity reduction with minimal BER degradation. Magnitude pruning (MP) is adopted as a reference technique based on prior work [9].

Experimental setup

Figure 2 illustrates the end-to-end experimental setup of a high-speed IM/DD optical link as a unified system diagram with labelled sections (a)–(e), used to evaluate advanced DSP under practical hardware constraints. In Fig. 2(a), the transmission chain is shown: offline TX DSP generates PAM signals with pulse shaping, converted by a high-speed AWG (Keysight M8199A, 256 GS/s), amplified, and applied to an MZM driven by a 1550.03 nm CW laser with polarization control (PC). The signal is amplified by an EDFA, while a VOA controls launch power. A 1% optical tap (PM) monitors the signal, while 99% passes through a 4.2-km SMF channel. At the receiver, a second VOA controls received optical power before the photodiode (PD), with an additional 1% tap for monitoring. The signal is detected by a photodiode (PD), amplified by a transimpedance amplifier (TIA), captured by a real-time oscilloscope (RTS, 80 GS/s), and processed offline using RX DSP, including synchronization, equalization (7% training, remainder for testing), and BER evaluation.

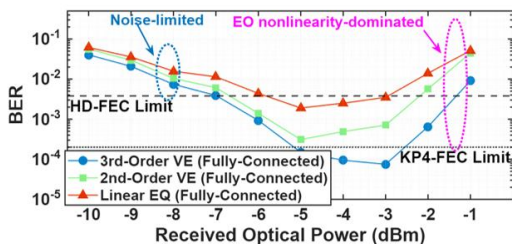


Fig. 3. BER versus received optical power over 4.2-km SMF.

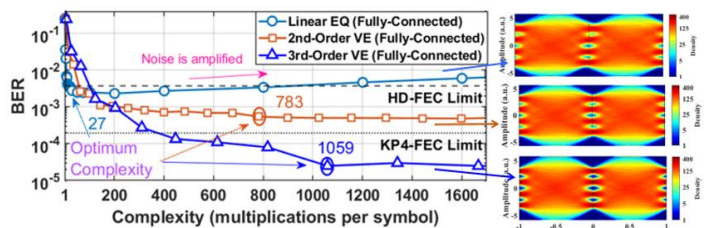


Fig. 4. BER-complexity trade-off at -3 dBm over a 4.2-km SMF link

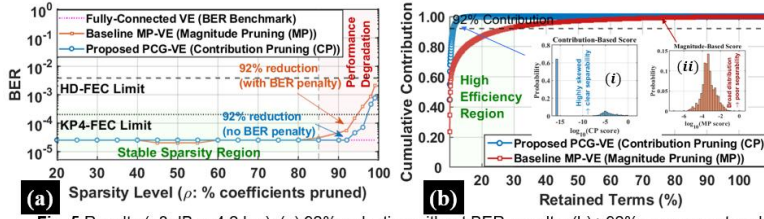


Fig. 5 Results (-3 dBm, 4.2 km): (a) 92% reduction without BER penalty; (b) $>92\%$ energy captured by CP; (i) MP shows poor separability; (ii) CP enables clear separability and efficient pruning.

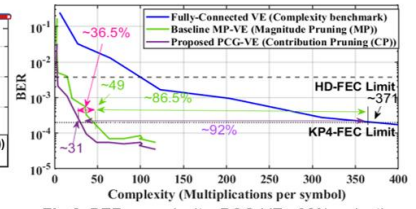


Fig. 6. BER-complexity: PCG-VE $\sim 92\%$ reduction ($\sim 37\%$ vs. MP-VE) with near KP4-FEC.

The remaining labelled sections highlight signal characteristics: Fig. 2(b) shows the optical spectrum, Fig. 2(c) presents the received signal before equalization, Fig. 2(d) illustrates the signal after linear equalization at ROP = -3 dBm, and Fig. 2(e) shows the improved signal after nonlinear equalization at the same power. This setup captures the effects of bandwidth limitation, noise, and electro-optic nonlinearity, providing a realistic platform for validating advanced DSP techniques in high-speed optical transmission systems.

Results and discussion

Figure 3 reveals the BER versus Received Optical Power (ROP) for linear EQ and fully-connected 2nd- and 3rd-order VEs (27/27/10 taps) over a 4.2-km SMF link. At low ROP (-10 to -7 dBm), the system operates in a noise-limited regime, where all equalizers exhibit high BER due to insufficient SNR. As the ROP increases, BER improves and Volterra equalizers outperform linear EQ by effectively compensating residual distortion. At higher ROP levels (-3 to -1 dBm), performance degrades due to electro-optic nonlinearity, leading to a nonlinear-dominated regime. The 3rd-order VE consistently outperforms the 2nd-order VE, achieving the best performance and maintaining BER below the KP4-FEC threshold over a wider power range.

Figure 4 shows the BER-complexity trade-off for fully connected linear EQ, 2nd-order VE, and 3rd-order VE at ROP = -3 dBm over a 4.2-km SMF link. Increasing complexity improves BER for VEs, with the 3rd-order VE achieving the best performance below the KP4-FEC limit, but at significantly higher computational cost. The optimal operating points are highlighted (27, 783, and 1059 multiplications), where each equalizer attains minimum BER. Beyond these points, further complexity provides diminishing returns, while linear EQ suffers performance degradation due to noise enhancement. This reveals that most Volterra terms are redundant, contributing little to performance while increasing complexity, thereby motivating the proposed approach to retain only dominant, physically meaningful terms for near-optimal BER with substantial complexity reduction.

Figure 5 illustrates the performance and underlying mechanism of the proposed pruning approach at a ROP of -3 dBm over a 4.2-km SMF link using fully-connected VEs (27/27/10 taps). As shown in Fig. 5(a), the BER versus sparsity level shows that the proposed PCG-VE (Contribution

Pruning, CP) maintains stable performance across a wide sparsity range and achieves up to 92% complexity reduction without BER penalty, whereas baseline MP-VE (Magnitude Pruning, MP) suffers rapid degradation at high sparsity level due to loss of important terms. From Fig. 5(b), the cumulative contribution shows that CP captures over 92% of the signal energy with only a small fraction of retained terms, indicating strong energy compaction, while MP requires significantly more terms to reach the same level. From Fig. 5(i), the Magnitude-based score distribution is broad and smooth, indicating poor separability between important and negligible terms, while in Fig. 5(ii), the contribution-based score distribution is highly skewed, clearly separating dominant and insignificant terms. These results demonstrate that CP effectively identifies physically relevant Volterra components, enabling efficient complexity reduction while preserving BER performance.

Figure 6 demonstrates BER versus complexity (multiplications per symbol) for fully connected VE, baseline MP-VE, and proposed PCG-VE, where complexity scales with retained coefficients. Increasing coefficients improves BER but rapidly increases complexity. The full VE achieves the best performance (~ 371 multiplications at the KP4-FEC limit) and serves as the benchmark. baseline MP-VE reduces complexity ($\sim 86.5\%$) via magnitude pruning but degrades BER due to loss of important nonlinear terms. In contrast, proposed PCG-VE retains only contribution-relevant coefficients, achieving $\sim 92\%$ complexity reduction and $\sim 36.5\%$ lower complexity than baseline MP-VE while maintaining near-optimal BER at the KP4-FEC limit, enabling a more efficient design.

Conclusion

In conclusion, we demonstrate a physics-aware contribution-guided Volterra equalization framework (PCG-VE) enabling efficient nonlinear compensation in a 100-Gb/s PAM4 IM/DD link. The proposed PCG-VE achieves BER comparable to the fully connected VE with $\sim 92\%$ coefficient reduction and no penalty, while maintaining KP4-FEC compliance even at 96% reduction, significantly outperforming conventional pruning methods. The enhanced robustness and extended FEC-compliant region establish PCG-VE as a promising low-complexity solution for next-generation IM/DD systems.

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